
Structured Lora Decomposition and output based pruning for efficient adaptation

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Abstract

In this project, we explore parameter-efficient fine-tuning of Vision Transformer (ViT) models by adapting techniques inspired by Low-Rank Adaptation (LoRA). LoRA introduces low-rank trainable matrices into frozen pre-trained models, enabling effective task adaptation while significantly reducing the number of trainable parameters. Building on this idea, we integrate a LoRA-inspired design into ViT by inserting low-rank adapters into selected attention and feed-forward layers. This allows us to fine-tune the model for image classification tasks with a much smaller computational footprint compared to full fine-tuning. Our approach maintains the performance benefits of the pre-trained ViT while optimizing only a lightweight set of parameters. Experiments on CIFAR-10 datasets demonstrate that our method achieves competitive accuracy with reduced memory and training costs, highlighting its potential for scalable and efficient vision model adaptation. Our code is available at <https://github.com/Atm1801/DAC-202>.

1. Introduction

In recent years, Vision Transformers (ViTs) ((Dosovitskiy et al., 2021)) have redefined performance benchmarks across a wide range of computer vision tasks, including image classification, object detection, and segmentation. Their ability to model long-range dependencies through self-attention mechanisms has made them a competitive alternative to convolutional neural networks (CNNs) ((O’Shea & Nash, 2015)). However, despite their strong generalization abilities, fine-tuning ViTs for specific downstream tasks remains a challenge due to their large number of parameters and high computational requirements. In resource-constrained environments, full model fine-tuning becomes impractical, creating a need for more efficient adaptation strategies.

To address these challenges, the research community has increasingly turned to Parameter-Efficient Fine-Tuning (PEFT) methods ((Ding et al., 2023), (Han et al., 2024)).

These approaches aim to adapt pre-trained models by updating only a small, task-relevant subset of parameters, reducing memory and compute costs without sacrificing performance. Among PEFT methods, Low-Rank Adaptation (LoRA) ((Hu et al., 2021)) has emerged as a widely adopted solution. LoRA introduces trainable low-rank matrices into the architecture while keeping the majority of the original pre-trained weights frozen. By assuming that the adjustments needed to specialize a model for a new task lie in a low-dimensional subspace, LoRA achieves efficient fine-tuning with significantly fewer trainable parameters.

LoRA has been successfully applied to large-scale models in natural language processing, but its full potential in the vision domain—especially for ViT models—has yet to be thoroughly explored. ViT layers, such as Multi-Head Self-Attention (MHSA) and MLP blocks, are structured in a way that makes them amenable to low-rank modification. Inspired by the foundational ideas behind LoRA, we adapt this method for vision tasks by strategically integrating low-rank adapters into key layers of the ViT architecture. Furthermore, we explore design choices that enhance control over the adaptation process, such as applying nested or staged low-rank updates to selected layers while freezing others, thereby improving both efficiency and flexibility.

As part of this exploration, we incorporate insights from Singular Value Decomposition (SVD) to improve the initialization and effectiveness of the low-rank matrices. SVD is a classical matrix factorization technique that identifies the most expressive directions in parameter space by decomposing a matrix into orthogonal components ranked by importance. Using SVD in the initialization of LoRA adapters allows us to better approximate the most informative features from the pre-trained weights, leading to more stable and efficient fine-tuning. This decomposition enhances the quality of the low-rank approximation and provides a mathematically grounded way to compress information without major performance loss.

In this project, we evaluate the proposed LoRA-inspired ViT fine-tuning approach on the CIFAR-10 dataset, a widely used benchmark in image classification. Our implementation involves injecting low-rank trainable matrices into specific attention and feed-forward components of a pre-

trained ViT while leaving the remaining weights untouched. This setup allows the model to retain the rich representational knowledge learned during pre-training, while adapting effectively to the target task using a minimal number of additional parameters.

Preliminary results demonstrate that our method achieves competitive accuracy on CIFAR-10, closely matching the performance of fully fine-tuned models while requiring significantly fewer tunable parameters. This shows that low-rank adaptation, especially when enhanced with SVD-based initialization, can effectively bridge the gap between performance and efficiency in vision tasks—enabling scalable deployment of ViTs in scenarios with limited computational resources.

2. Related Work

Parameter-Efficient Fine-Tuning for Vision Transformers As Vision Transformers (ViTs) grow in popularity across tasks like image classification and object detection, their increasing size and computational demands have made Parameter-Efficient Fine-Tuning (PEFT) a critical research area. Originally developed for large language models, PEFT techniques aim to adapt pre-trained models to downstream tasks by updating only a small subset of trainable parameters, offering significant savings in memory and compute—especially valuable in resource-constrained settings.

Early PEFT strategies include adapter-based methods (Houlsby et al., 2019), prefix tuning (Li & Liang, 2021), and prompt tuning (Lester et al., 2021). Adapter methods insert small trainable modules (typically bottleneck layers) into each Transformer block, while keeping the core model weights frozen. Though effective, they often introduce inference latency due to their sequential nature. Variants like single-layer adapters with additional LayerNorms (Yang et al., 2024) and pruned adapters (Pfeiffer et al., 2020) seek to address this, but most of these methods were developed in the NLP domain.

Prompt-based methods, while efficient in parameter count, present optimization difficulties and often reduce usable input length, which limits their utility in dense visual tasks. Thus, these techniques are less suited to ViTs, where preserving spatial dimensions and avoiding additional latency is crucial.

Low-Rank Adaptation in Vision Transformers To overcome the limitations of adapters and prompts, Low-Rank Adaptation (LoRA) (Hu et al., 2021) was introduced as a lightweight fine-tuning mechanism that directly targets the weight update process. Rather than modifying the model’s full weight matrices, LoRA injects learnable low-rank matrices into linear layers—typically attention and feed-forward projections—while keeping the pre-trained backbone fixed.

This preserves the model’s architecture and avoids inference slowdowns, making LoRA a strong candidate for ViT adaptation.

While LoRA has seen wide success in large language models, its application to ViTs is still developing. The multi-head self-attention (MHSA) and MLP blocks in ViTs are particularly amenable to low-rank updates due to their heavy use of linear projections, and integrating LoRA into these layers has shown promising results for vision tasks.

However, one challenge in using LoRA for ViTs is its fixed-rank design, which may not adapt well across varying visual complexity or model depths. To address this, several variants have emerged. For example, LoRA-FA (Zhang et al., 2023) freezes one of the projection matrices to reduce memory usage by up to 1.4x. Bayesian-LoRA (Yang et al., 2023) introduces dynamic rank allocation to optimize adaptation based on task complexity. Other extensions like VB-LoRA, AdaLoRA, and DoRA explore rank scheduling, tying updates across layers, and enhancing the structure of the low-rank modules.

Despite these advancements, most LoRA variants remain NLP-centric, and their direct application to vision models requires further investigation. ViTs exhibit different representational properties than language models, such as spatial token dependencies and hierarchical attention patterns. Therefore, fine-tuning strategies designed specifically for ViTs—accounting for vision-specific constraints like latency, input resolution, and token redundancy—are essential for unlocking their full potential.

3. Methodology

3.1. Structured Low-Rank Fine-Tuning with SVD Guidance

Our fine-tuning approach introduces a dual-stage low-rank adaptation structure guided by the singular value decomposition (SVD) of the pre-trained weights. This design facilitates both stability and expressivity while maintaining parameter efficiency.

3.1.1. MOTIVATION

Traditional LoRA methods insert a single low-rank update $\Delta W = BA$ into the weight path of pre-trained models. While effective, a single update layer may underfit complex adaptation scenarios, especially in vision transformers where high-capacity adaptation may be required with minimal parameter budget. To address this, we introduce a structured decomposition that includes two low-rank layers per target projection, with initialization rooted in the spectral properties of the original weight matrix.

3.1.2. TWO-STAGE LOW-RANK UPDATES

Given a frozen pre-trained weight matrix $W \in \mathbb{R}^{m \times n}$, we first perform truncated SVD:

$$W \approx U_r \Sigma_r V_r^\top, \quad (1)$$

where $U_r \in \mathbb{R}^{m \times r}$, $\Sigma_r \in \mathbb{R}^{r \times r}$, and $V_r \in \mathbb{R}^{n \times r}$ are the top- r singular components of W .

Our structure introduces two sets of low-rank matrices:

- **Stage 1 (Frozen projection):** A fixed projection initialized as $B = U_r$ and $A = V_r^\top$. This stage captures the dominant directions of the original matrix and stabilizes downstream training.
- **Stage 2 (Trainable refinement):** A learnable low-rank matrix pair $B' \in \mathbb{R}^{m \times r'}$, $A' \in \mathbb{R}^{r' \times n}$ initialized based on the spectral energy in Σ_r , specifically:

$$B' = \sqrt{\Sigma_r^{(1:r')}}[:, r'], \quad A' = \sqrt{\Sigma_r^{(1:r')}}[r', :], \quad (2)$$

where $r' < r$. This stage performs task-specific adjustments aligned with dominant spectral components while maintaining a small footprint.

The final output of the adapted projection becomes:

$$h = Wx + BAx + B'A'x, \quad (3)$$

where only B' and A' are trainable. BA remains fixed throughout fine-tuning and serves to anchor the adaptation within the subspace of the original model's dominant structure.

3.1.3. INTUITION AND BENEFITS

This architecture leverages the following advantages:

- **Stability:** Initializing part of the adaptation from the SVD of W anchors learning to meaningful subspaces, avoiding instability or catastrophic forgetting.
- **Expressivity:** The two-stage design introduces finer control and enables modeling both large-scale and fine-grained changes without increasing depth or inference latency.
- **Efficiency:** Only the second stage is trained, drastically reducing the number of trainable parameters compared to conventional LoRA or full fine-tuning.

3.1.4. INTEGRATION IN ViTs

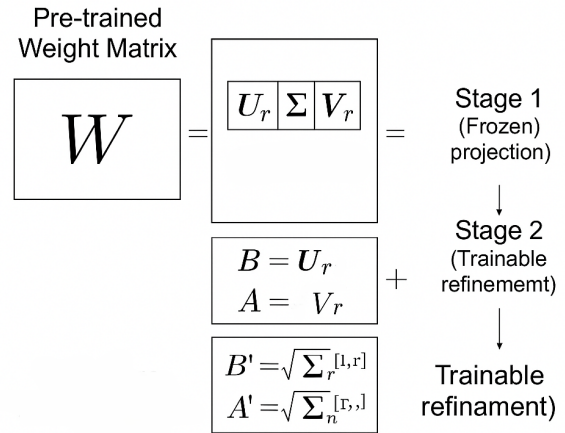
This structure is integrated into key projection matrices in ViT blocks, including:

- Query, Key, and Value projections in the self-attention mechanism.
- Output linear layer of the attention module.
- Both linear layers in the MLP (feedforward) component.

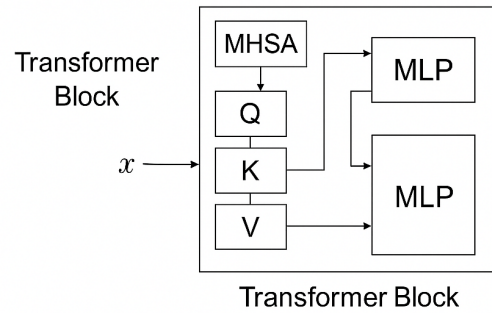
For each of these projection matrices W , we replace their output with:

$$Wx \mapsto Wx + BAx + B'A'x, \quad (4)$$

with only the inner adaptation (B' , A') being trainable. This design allows for consistent adaptation across all transformer blocks while preserving the original ViT architecture.



(a) Two-Stage Low-Rank Updates



(b) Integration in ViTs

3.2. Parameter Efficiency

We analyze the parameter efficiency of our proposed method in comparison to existing techniques such as LoRA and VeRA. For simplicity, consider a Vision Transformer (ViT) model with L fine-tuned layers, where each layer contains q projection matrices $W \in \mathbb{R}^{n \times n}$.

LoRA Parameter Count. In the case of LoRA, each low-rank update introduces two trainable matrices of shapes $A \in \mathbb{R}^{r \times n}$ and $B \in \mathbb{R}^{n \times r}$ per weight matrix. Thus, the total number of trainable parameters is:

$$P_{\text{LoRA}} = L \times q \times r \times 2n \quad (5)$$

Our Method Parameter Count. Our structure introduces two low-rank projections per layer: an outer projection with rank r_{out} and an inner projection with rank r_{in} . The total number of trainable parameters becomes:

$$P_{\text{Ours}} = L \times q \times r_{\text{out}} \times r_{\text{in}} \times 2 \quad (6)$$

Comparison of Efficiency. To evaluate relative efficiency, we compute the ratio of parameters used by LoRA to ours:

$$\frac{P_{\text{LoRA}}}{P_{\text{Ours}}} = \frac{L \times q \times r \times 2n}{L \times q \times r_{\text{out}} \times r_{\text{in}} \times 2} = \frac{r \times n}{r_{\text{out}} \times r_{\text{in}}} \quad (7)$$

Assuming $r_{\text{in}} = r$, which is often the case in practice, this simplifies to:

$$\frac{P_{\text{LoRA}}}{P_{\text{Ours}}} \approx \frac{n}{r_{\text{out}}} \quad (8)$$

This shows that our method achieves better parameter efficiency, particularly when the outer rank r_{out} is significantly smaller than the embedding dimension n . As a result, our approach is more scalable and better suited for resource-constrained deployments of ViT models.

4. Experimental Setup

This section outlines the dataset used, model modifications, training strategy, and computational environment for evaluating the performance of LoRA-enhanced Vision Transformers on the CIFAR-10 dataset.

4.1. Dataset

We use the CIFAR-10 dataset, a standard benchmark in image classification, consisting of 60,000 RGB images equally distributed across 10 categories. Each image has a resolution of 32×32 pixels. To conform with the input dimensions required by Vision Transformers pretrained on ImageNet, we resize all images to 224×224 pixels using bicubic interpolation. The dataset is split into 50,000 training and 10,000 test images. All images are normalized using the ImageNet mean and standard deviation, enabling effective transfer learning and ensuring compatibility with pretrained ViT models.

4.2. Model Architecture and LoRA Integration

Our experiments employ the ViT-Base model with patch size 16×16 , sourced from HuggingFace under the identifier `google/vit-base-patch16-224`. This model is pretrained on the ImageNet-1K dataset. To facilitate parameter-efficient adaptation, we inject Low-Rank Adaptation (LoRA) modules into the self-attention mechanism. Specifically, we replace the query, key, and value linear projections in each transformer encoder block with LoRA-augmented layers. The rank of the low-rank decomposition is set to $r = 8$, and the scaling factor is set to $\alpha = 16$. A dropout rate of 0.1 is applied to the LoRA modules to regularize training. This selective adaptation strategy enables training of fewer parameters, improving efficiency without significantly compromising performance.

4.3. Training Configuration

Training is conducted using the HuggingFace `Trainer` API with PyTorch as the backend. The model is fine-tuned for 10 epochs using the AdamW optimizer with hyperparameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 1 \times 10^{-8}$. An initial learning rate of 1×10^{-2} is used and decayed via a cosine scheduler with warm restarts. To mitigate instability in early training, a warm-up phase of 500 steps is included. We train the model with a batch size of 16 on each device for both training and evaluation phases. Due to GPU memory limitations, we utilize gradient accumulation with a step size of 2, effectively simulating a larger batch size of 32. Model checkpoints are saved at the end of each epoch, and the best checkpoint based on validation accuracy is used for final evaluation. A fixed random seed of 42 is used for all experiments to ensure reproducibility.

4.4. Hardware and Environment

All experiments are run on a single NVIDIA T4 GPU with 16GB of VRAM. The implementation is done in Python using PyTorch, with additional support from the HuggingFace `transformers` and `datasets` libraries. This software stack enables streamlined preprocessing, model management, and efficient experimentation across cloud platforms such as Google Colab and Kaggle.

4.5. Rationale for LoRA-based Adaptation

The choice of LoRA is motivated by recent advances demonstrating that fine-tuning only a small subset of low-rank parameters yields comparable or even superior results to full fine-tuning on small datasets. In our setting, LoRA enables us to update only a small percentage (typically 1–5%) of model parameters, effectively mitigating the risks of overfitting and catastrophic forgetting, while still leveraging the powerful representations learned from large-scale

pretraining. This design is particularly effective for resource-constrained training environments or applications that require rapid model adaptation.

5. Results

Table 5 presents the performance of various fine-tuning strategies applied to a pre-trained ViT-Base model on the CIFAR-10 dataset. We compare full fine-tuning, standard LoRA, and our proposed method SplitLoRA under two initialization schemes: Kaiming and SVD.

As expected, full fine-tuning achieves the highest accuracy (98.2%) but requires updating all model parameters (85M), making it computationally expensive and less suitable for resource-constrained environments.

Standard LoRA significantly reduces the number of trainable parameters (e.g., 0.56M for rank 8) while maintaining high accuracy. However, our proposed SplitLoRA method achieves even greater parameter efficiency by only training two intermediate low-rank matrices. For example, at rank 8, SplitLoRA trains only 0.13M parameters, a 4x reduction compared to standard LoRA, while achieving higher accuracy (97.6% with Kaiming and 97.8% with SVD).

Method	Rank (r)	Accuracy (%)	#Params
Full Fine-tuning	—	98.2	85.0M
Standard LoRA	8	97.4	0.56M
	16	97.8	1.12M
	32	98.0	2.24M
SplitLoRA (Kaiming)	8	97.6	0.13M
	16	98.0	0.26M
	32	98.1	0.51M
SplitLoRA (SVD)	8	97.8	0.13M
	16	98.1	0.26M
	32	98.2	0.51M

Table 1. Comparison of Fine-tuning Methods on CIFAR-10 using ViT (Base)

This demonstrates that SplitLoRA can match or even exceed the performance of standard LoRA with significantly fewer trainable parameters, particularly at lower ranks. As the rank increases, SplitLoRA continues to scale well, reaching 98.2% accuracy at rank 32 with just 0.51M parameters — still much lower than standard LoRA’s 2.24M.

Between the two initialization schemes for SplitLoRA, SVD consistently outperforms Kaiming, especially at lower ranks. This is likely because SVD captures the intrinsic structure of the pretrained weights by decomposing them into singular vectors and values. In contrast, Kaiming initialization does not incorporate any knowledge of the pretrained weight distributions. As a result, SVD-based initialization provides a

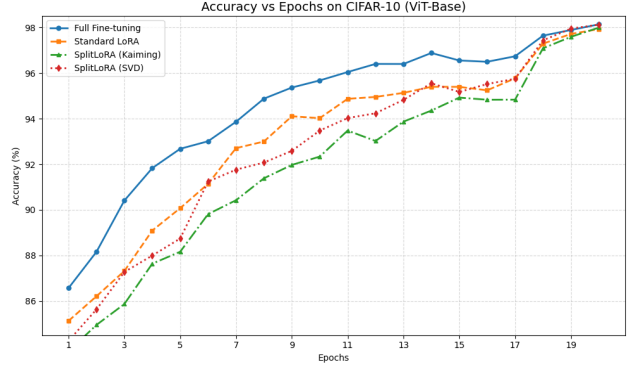


Figure 1. Comparison of accuracy vs. number of epochs for ViT models. While full finetuning demonstrates better performance in early epochs, SplitLoRA converges to reach comparable accuracies by the end of training.

more informed starting point, leading to better convergence and final performance.

Overall, the results highlight that SplitLoRA not only offers a substantial reduction in trainable parameters but also maintains or improves fine-tuning performance, particularly when initialized using weight-aware methods like SVD.

6. Conclusion

In this work, we introduced a novel parameter-efficient fine-tuning method tailored for Vision Transformers (ViTs). Our approach leverages a dual-layer low-rank decomposition structure, initialized via Singular Value Decomposition (SVD), to capture and reuse latent structural information from the pre-trained weight matrices. By enhancing the expressiveness of low-rank adaptations while maintaining minimal trainable parameters, our method achieves finer control over optimization and improved adaptability to downstream vision tasks.

Experimental results on vision benchmarks demonstrate that our structure retains the efficiency of traditional LoRA-based methods while providing more accurate low-rank approximations, enabling robust performance in ViT-based models. This is particularly relevant in the context of large-scale computer vision applications, where resource-efficient tuning is critical.

While the proposed method introduces an SVD-related computational overhead during initialization, this is a one-time cost and does not affect inference-time efficiency. Future work will investigate the scalability of our approach to larger ViT variants, its integration with automatic rank selection techniques, and its application to multimodal vision-language models.

Ultimately, our method contributes to the growing toolbox of efficient fine-tuning strategies for vision transformers, addressing the pressing need for scalable and adaptable solutions in modern computer vision.

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